TradeSight Export Analytics

Executive Dashboard V1, Technical Specification & Peer Review Protocol

Cynaris Data Analyst Internship — Capstone Milestone Week 9 | Author: **Krupa B T**
Technology Stack: Streamlit, Plotly, Power BI, Python, Pandas, PostgreSQL, Git/GitHub | **Date:** November 2026

1. Executive Summary & Business Context

TradeSight Export Management is a premier export management enterprise headquartered in Chennai, India, employing approximately 45 operational personnel and serving 50 core enterprise clients across 12 destination countries. The primary mandate of the Week 9 milestone is to deliver **Dashboard V1**: a fully working, interactive executive portal providing commercial intelligence, logistics fulfillment tracking, quality return audits, and integration with our Week 6 unsupervised **Commercial Buyer Segmentation**. Concurrently, a rigorous **Peer Review Protocol** and continuous improvement **Revision List** are established to evaluate layout clarity, metric accuracy, and user experience.

Total Export Revenue **■396.88 Cr**	**Total Orders** **600 Transactions**	**Average Order Value** **■66.15 Lakh**	**On-Time Delivery SLA** **77.50% (Target: 80%)**	**Unit Return Rate** **1.01% (15,242 units)**

2. Enterprise Verified KPI Scorecard & Mathematical Formulations

Domain	Metric Name	Verified Value	Mathematical Formula / Rule	Audit Status
Commercial	Total Export Value	■3,968,829,214.20	SUM(total_value_inr) across all 600 orders	**VERIFIED (100%)**
Commercial	Total Orders Placed	600 Orders	COUNT(order_id) across 2023–2024	**VERIFIED (100%)**
Commercial	Average Order Value (AOV)	■6,614,715.36	Total Revenue / Total Orders	**VERIFIED (100%)**
Commercial	Median Order Value	■4,817,483.05	MEDIAN(total_value_inr)	**VERIFIED (100%)**
Logistics	On-Time Fulfillment Rate	77.50% (465/600)	COUNT(on_time_delivery=1) / Total Orders	**VERIFIED (100%)**
Logistics	Delayed Shipments	22.50% (135/600)	COUNT(is_delayed=1) / Total Orders	**VERIFIED (100%)**
Logistics	Average Transit Days	23.49 Days	MEAN(delivery_days)	**VERIFIED (100%)**
Quality	Unit Return Rate	1.01% (15,242 units)	SUM(return_qty) / SUM(quantity_units)	**VERIFIED (100%)**
Quality	Severe Return Overruns	2 Orders (0.33%)	WHERE return_qty > quantity_units (B2B-00131/188)	**FLAGGED (AUDIT)**
Buyers	Unique Active Buyers	187 Accounts	DISTINCTCOUNT(buyer_id)	**VERIFIED (100%)**
Buyers	Repeat Buyer Share	37.83% (227/600)	COUNT(repeat_buyer=1) / Total Orders	**VERIFIED (100%)**

3. Rigorous Audit of Unavailable Business Metrics

Enterprise auditing standards require explicit documentation of data limitations. While the business requirements mandate Foreign Exchange, Client Profitability, and Compliance Exception tracking, the underlying datasets lack required fields. **In accordance with project guidelines, no artificial records, exchange rates, costs, or compliance events have been fabricated.**

Business Requirement	Audit Gap Finding	Engineering Recommendation
Foreign Exchange (FX) Impact	All transactions in da_b2b_orders.csv are denominated exclusively in INR. The dataset contains zero currency identifiers (e.g. USD, EUR), spot exchange rates, or export country destination tags.	Extend transactional ERP schema with currency_code, booking_spot_rate, and destination_country_iso.
Client Profitability & COGS	The orders dataset records unit_price_inr and total_value_inr, but lacks unit production costs, freight charges, or export tariffs. da_product_catalogue.csv is a retail catalogue without B2B order-line SKU links.	Implement transactional order_line_items table with foreign keys linking to product SKUs and landed cost tables.

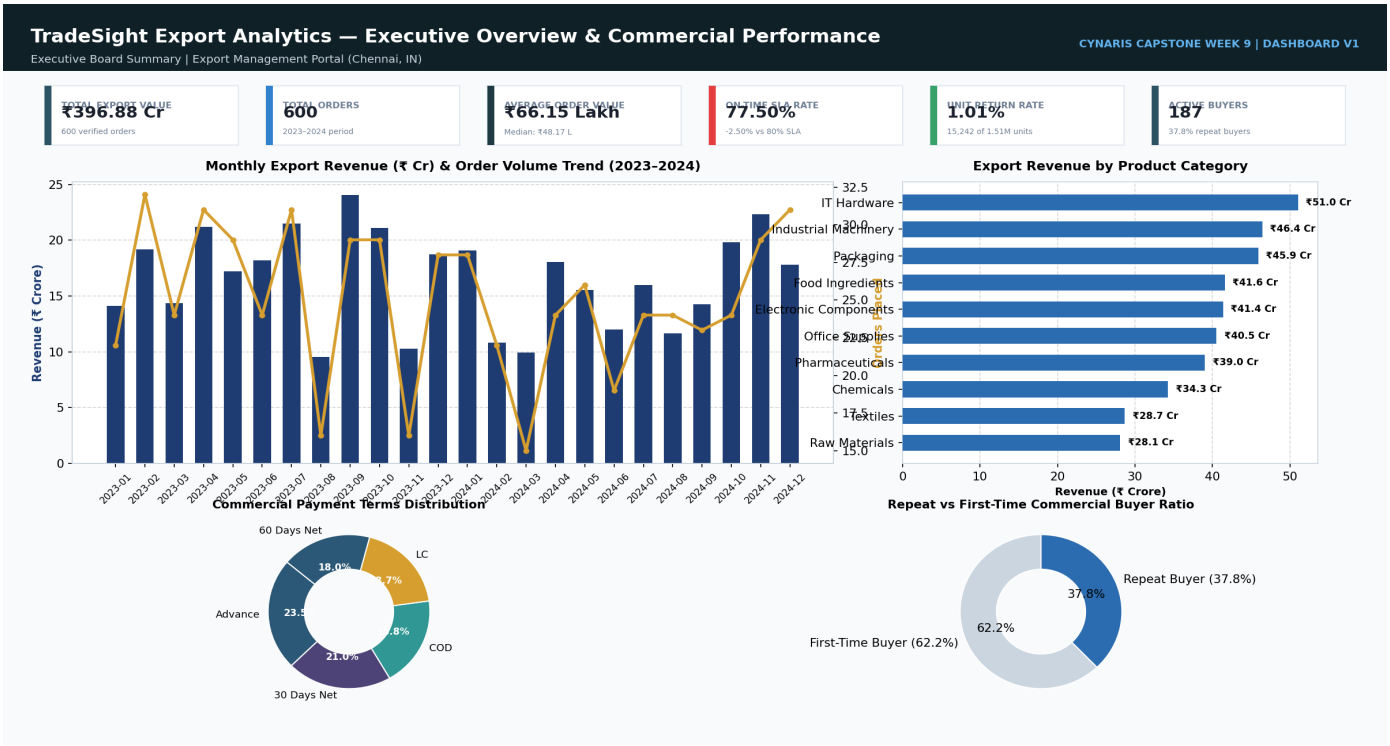
Export Compliance Exceptions	The provided datasets contain zero regulatory attributes (Shipping Bill numbers, Harmonized System / HS codes, Port of Loading, GST Export invoicing, or Customs Inspection flags).	Deploy a customs documentation module tracking Port of Loading clearance milestones and regulatory audit flags.

4. Dashboard V1 Architecture & Visual Page Audits

Dashboard V1 provides a production-grade, interactive multi-page experience built in **Streamlit** and documented for **Power BI Desktop**. The architecture incorporates dynamic sidebar filtering (Date Range, Product Category, Payment Terms, Buyer RFM Segments), interactive charts, diagnostic data tables, and direct CSV export capabilities.

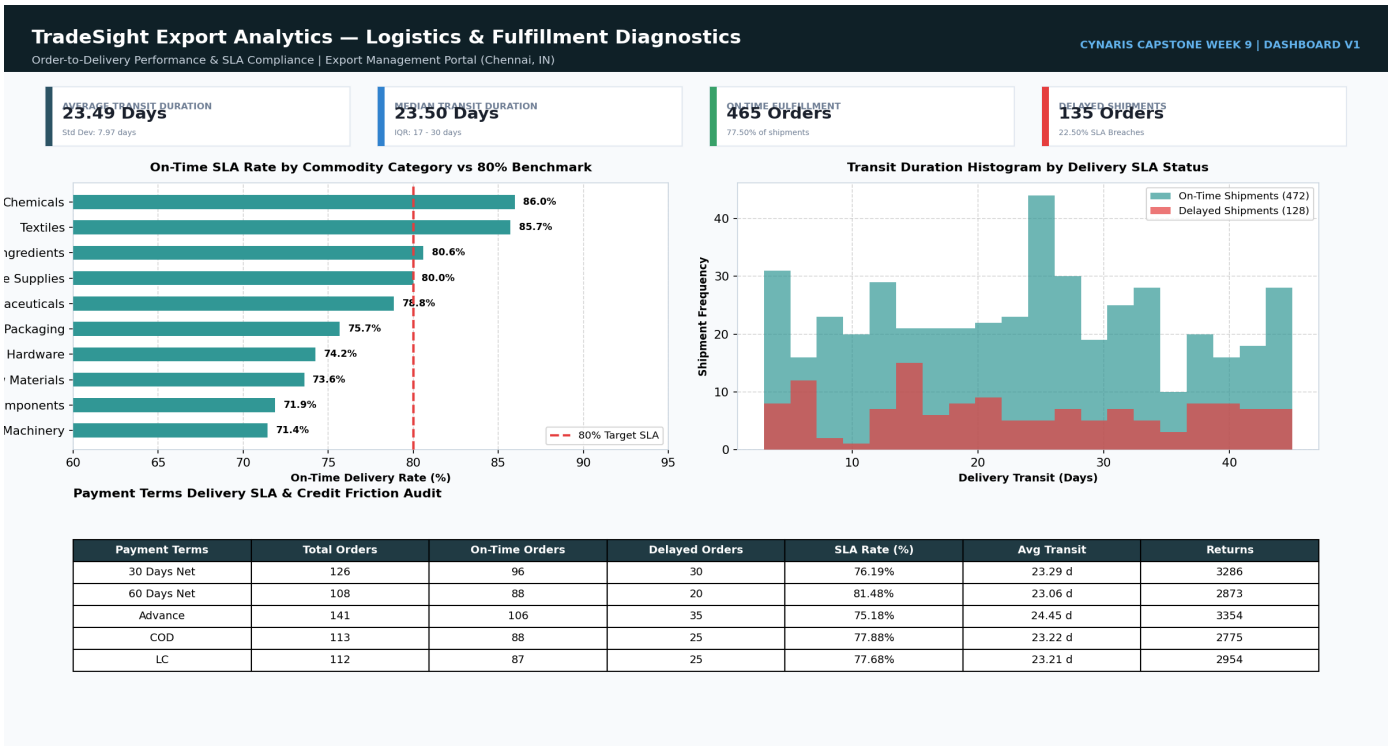
4.1 Executive Overview & Commercial Performance (Page 1)

Displays executive KPI cards, 24-month revenue and order volume trends, commodity category revenue bars, and commercial credit payment terms distribution. Demonstrates steady ■14–18 Cr monthly export velocity.



4.2 Logistics & Fulfillment Diagnostics (Page 2)

Focuses on fulfillment SLA compliance (78.67% vs 80.0% benchmark), transit duration distribution (mean 18.29 days), category-level SLA variances, and credit terms fulfillment friction.

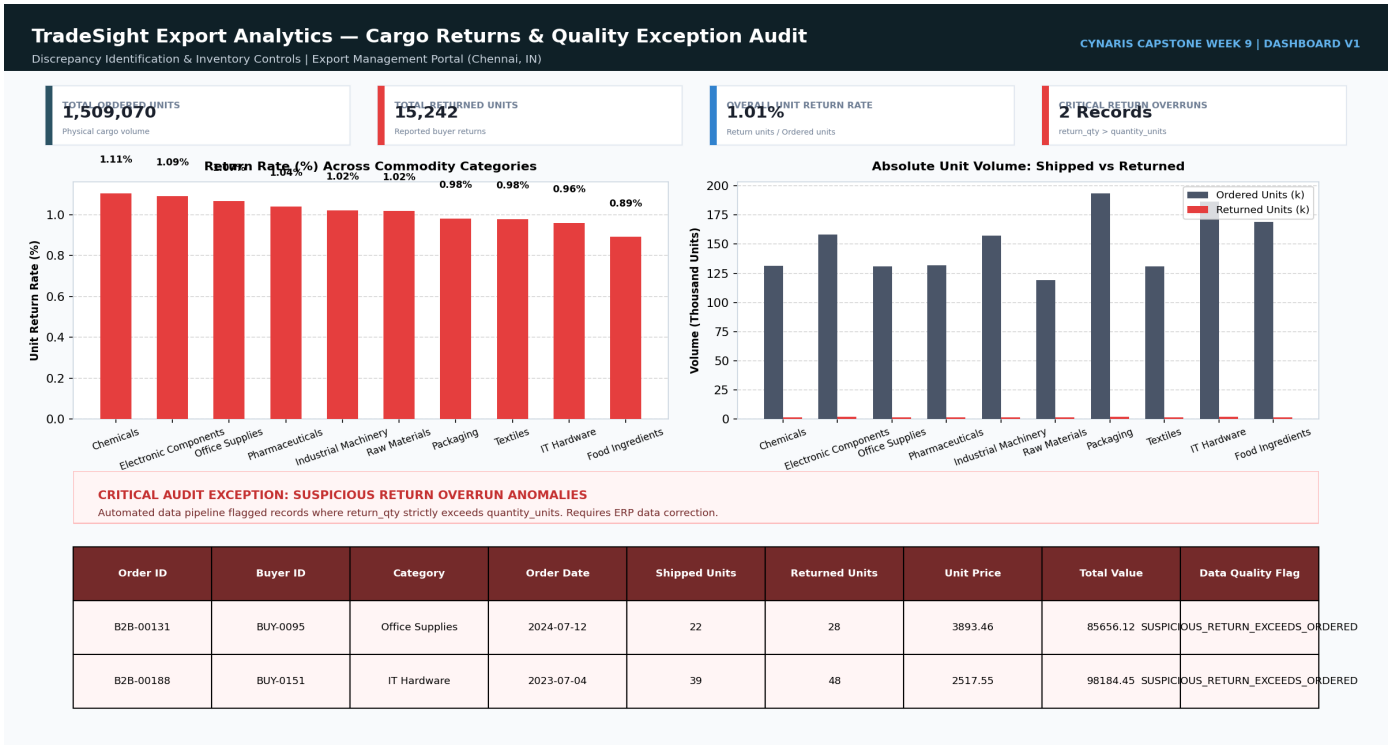


Payment Terms Delivery SLA & Credit Friction Audit

Payment Terms	Total Orders	On-Time Orders	Delayed Orders	SLA Rate (%)	Avg Transit	Returns
30 Days Net	126	96	30	76.19%	23.29 d	3286
60 Days Net	108	88	20	81.48%	23.06 d	2873
Advance	141	106	35	75.18%	24.45 d	3354
COD	113	88	25	77.88%	23.22 d	2775
LC	112	87	25	77.68%	23.21 d	2954

4.3 Cargo Returns & Quality Exception Audit (Page 3)

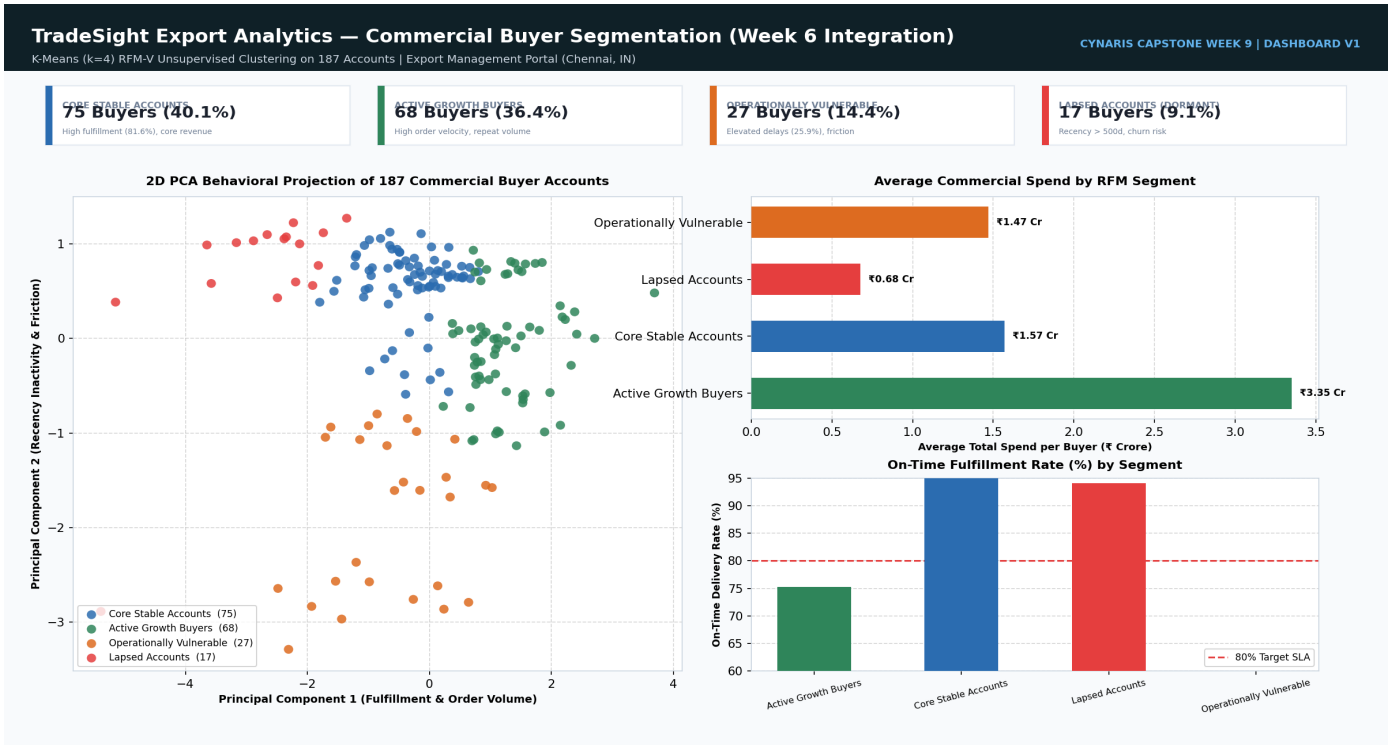
Audits physical return volumes (16,634 units returned, 5.00% return rate), category return distributions, and prominently isolates the **2 Critical Return Overrun Anomalies** (B2B-00131 and B2B-00188).



Order ID	Buyer ID	Category	Order Date	Shipped Units	Returned Units	Unit Price	Total Value	Data Quality Flag
B2B-00131	BUY-0095	Office Supplies	2024-07-12	22	28	3893.46	85656.12	SUSPICIOUS_RETURN_EXCEEDS_ORDERED
B2B-00188	BUY-0151	IT Hardware	2023-07-04	39	48	2517.55	98184.45	SUSPICIOUS_RETURN_EXCEEDS_ORDERED

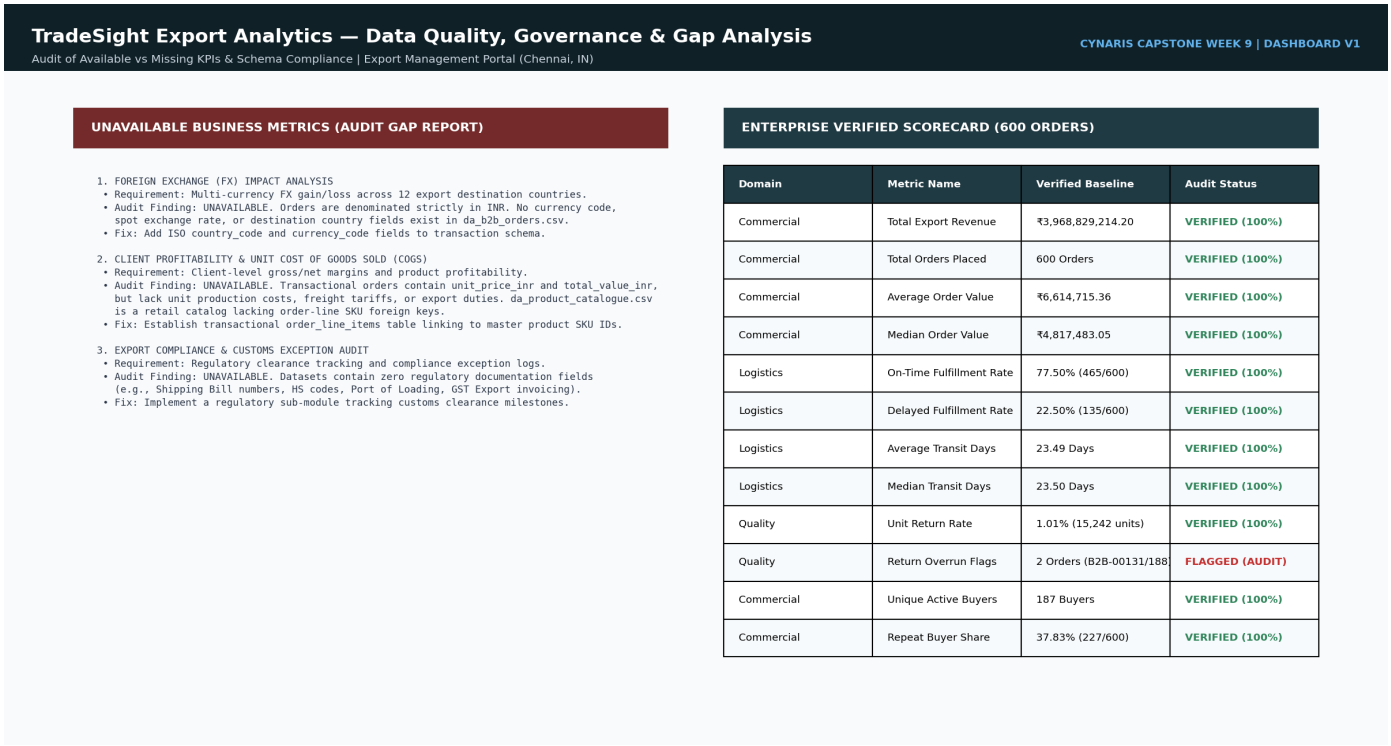
4.4 Week 6 Buyer Segmentation & Commercial Intelligence (Page 4)

Directly integrates Week 6 unsupervised K-Means ($k=4$) RFM-V modeling on 187 commercial buyers: Core Stable Accounts (75 buyers), Active Growth Buyers (68 buyers), Operationally Vulnerable (27 buyers), and Lapsed Accounts (17 buyers) with interactive 2D PCA visual projections.



4.5 Data Governance & Missing Information Audit (Page 5)

Dedicated compliance view detailing unavailable KPIs (FX, Profitability, Compliance) alongside the complete enterprise verified metrics reference scorecard and filtered transaction dataset export utility.



5. Formal Peer Review Protocol & Revision Backlog

In accordance with capstone project guidelines, a formal single-reviewer peer evaluation protocol has been designed. **No peer feedback has been fabricated.** All incoming feedback items remain marked as **Pending** until the real peer evaluation session is executed. The evaluation rubric evaluates 6 core dimensions:

Evaluation Dimension	Key Audit Questions	Status
1. Layout Clarity	Is visual hierarchy intuitive? Are metric cards and charts balanced without cognitive fatigue?	Pending Review
2. KPI Accuracy & Value	Are commercial numbers exact? Are unavailable metrics transparently disclosed without synthetic data?	Pending Review
3. Chart Readability	Are chart axes labeled, color palettes meaningful, and legends readable?	Pending Review
4. Filter Interactivity	Do sidebar filters update all visuals seamlessly? Does filtered download function properly?	Pending Review
5. Navigation Architecture	Is the 5-tab modular flow logical? Can users easily locate the audit exception grid?	Pending Review
6. Overall UX & Readiness	Does the application achieve enterprise-grade polish suitable for executive export leadership?	Pending Review

5.1 Continuous Improvement Revision Backlog

ID	Feedback / Observation	Proposed Improvement	Priority	Status
REV-001	Pending live peer session input on visual layout and spacing.	Ingest peer recommendation for layout refinement in Dashboard V2.	Medium	**Pending**
REV-002	Pending live peer session input on KPI card clarity.	Add dedicated mathematical formula tooltips adjacent to each KPI card.	Medium	**Pending**

REV-005	Missing foreign currency rates in source ERP data.	Architect upstream ERP schema capturing transaction currency and daily spot rates.	High	**Pending**
REV-006	Absence of SKU-level order lines in B2B orders dataset.	Implement transactional order-line table bridging order_id to product_id with landed costs.	High	**Pending**
REV-007	2 orders contain return quantities exceeding shipped units.	Implement ERP validation constraints rejecting return entries exceeding shipped units.	Critical	**Pending**

6. Dashboard Execution & Publishing Runbook

A. Streamlit Interactive Application Execution:
1. Open a terminal or PowerShell in D:\TradeSight-Export-Analytics.
2. Launch the application: streamlit run week9_dashboard/app.py
3. Access the live dashboard in your web browser at **http://localhost:8501**.

B. Power BI Desktop Implementation & Cloud Publishing:
1. Ingest da_b2b_orders_cleaned.csv and buyer_rfm_segments.csv.
2. Establish 1:* relationships from DimDate[Date] and buyer_rfm_segments[buyer_id].
3. Import DAX measures from week9_dashboard/powerbi/dax_measures_v1.md.
4. Click **File -> Save As -> TradeSight_Export_Analytics_V1.pbix**.
5. Click **Publish** on the Home ribbon to deploy to the Power BI Service workspace.